Math 2550 S01-06 Quiz 2

Section:
Name:
Student Number:

(1) Solve the initial value problem for $\mathbf{r}$ as a vector function of t.
Differential equation: $\frac{d^2\mathbf{r}}{dt^2} = e^t\mathbf{i} - e^{-t}\mathbf{j} + 4e^{2t}\mathbf{k}$
Initial conditions: $\mathbf{r}(0) = 3\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\frac{d\mathbf{r}}{dt}|_{t=0} = -\mathbf{i} + 4\mathbf{j}$

(2) Find the value of t so that the tangent line to

$$\mathbf{r}(t) = -t\mathbf{i} + t^2\mathbf{j} + (\ln t)\mathbf{k}$$

contains the point $(2, -5, -3)$. (Write down the procedure and there is no need to solve the resulted system of equations.)